

A PEAK-POINT PARTIAL ANSWER FOR ORDER-PRESERVING ISOMETRY EXTENSIONS

AGENT LANE 16

STATUS

This is a candidate partial result for the final Problem in Tanmoy Paul and T. S. S. R. K. Rao, *Order-preserving unique Hahn-Banach extensions*, arXiv:2504.03386 [1]. The result addresses the literal reading in which an order-preserving isometric embedding

$$\Phi : A(K) \longrightarrow C(K)$$

is to have a unique positive linear isometric extension

$$T : C(K) \longrightarrow C(K).$$

It does not settle the broader projection-valued or operator-space variants suggested by the surrounding paragraph in the source paper.

SOURCE PROBLEM

The source problem appears on PDF page 14, in the final displayed Problem of [1]. A crop of the statement is included below.

Problem 4.8. *Operator theoretic versions of the results considered here are still open. It is a folklore result that for an infinite dimensional Choquet simplex K , whose extreme boundary is not closed, the identity map $i : A(K) \rightarrow C(K)$ has no norm preserving extension to $C(K)$. To see this, note that existence of an extension implies that $A(K)$ is the range of a contractive projection on $C(K)$. It follows from Theorem 5 in Section 10 of [9] (see also the classification scheme in [11]), since $A(K)$ is the range of a contractive projection in a space of continuous functions, $\partial_e A(K)_1^* \cup \{0\}$ is a weak*-compact set. Since $\mathbf{1} \in A(K)$, 0 can not be an accumulation point. Now it is easy to see that $\partial_e K$ is a closed set, giving the required contradiction. In view of the operator versions from [15], it would be interesting to find conditions under which an order-preserving isometric embedding $\Phi : A(K) \rightarrow C(K)$ has unique extension to an into order-preserving isometry from $C(K)$ to $C(K)$?*

In transcription, the final sentence asks:

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In view of the operator versions from Rao, it would be interesting to find conditions under which an order-preserving isometric embedding $\Phi : A(K) \rightarrow C(K)$ has unique extension to an into order-preserving isometry from $C(K)$ to $C(K)$.

DEFINITIONS

Throughout, K is a compact convex subset of a locally convex real topological vector space, $A(K)$ denotes the real Banach space of continuous affine real-valued functions on K , and $C(K)$ denotes the real Banach lattice of continuous real-valued functions on K , both with the supremum norm.

Definition 1. A point $x \in K$ is a peak point for $C(K)$ if there is $p_x \in C(K)$ such that $0 \leq p_x \leq 1$, $p_x(x) = 1$, and $p_x(y) < 1$ for every $y \neq x$.

Every compact metrizable Hausdorff space has this property: for example, if d is a compatible metric, $p_x(y) = 1 - \min\{d(x, y), 1\}/2$ peaks at x .

PROOF INTUITION

Let $T : C(K) \rightarrow C(K)$ be a positive unital extension of a composition embedding $a \mapsto a \circ h$. Evaluating Tf at a point s gives, by the Riesz-Markov representation theorem, integration against a probability measure μ_s on K . Because T extends $a \mapsto a \circ h$, the measure μ_s represents the point $h(s)$ on affine functions. The isometry condition then forces these representing measures to be point masses: if a continuous peak function at $h(s)$ has norm one, so does its image under T , and at a point where this norm is attained the associated probability measure must be supported entirely at $h(s)$. The affine separation of points and injectivity of h move that conclusion back to the original point s . Thus each $\mu_s = \delta_{h(s)}$, so T is necessarily the composition operator $f \mapsto f \circ h$.

The part of the source problem that remains open is the general classification of embeddings $\Phi : A(K) \rightarrow C(K)$ for which this uniqueness holds, especially under interpretations closer to the projection or operator-space language in the source paragraph.

THE PARTIAL THEOREM

Theorem 1. *Let K be a compact convex set such that every point of K is a peak point for $C(K)$. Let $h : K \rightarrow K$ be a homeomorphism, and define*

$$\Phi : A(K) \longrightarrow C(K), \quad \Phi(a) = a \circ h.$$

Then Φ is a unital order-preserving linear isometry. Moreover, it has a unique positive linear isometric extension $T : C(K) \rightarrow C(K)$, namely

$$T(f) = f \circ h \quad (f \in C(K)).$$

Proof. Since h is onto, the map $a \mapsto a \circ h$ preserves suprema of absolute values and hence is an isometry from $A(K)$ into $C(K)$. It is linear, unital, and order preserving. The composition operator

$$C_h : C(K) \rightarrow C(K), \quad C_h(f) = f \circ h,$$

is a positive linear isometry extending Φ , so existence is clear.

It remains to prove uniqueness. Let $T : C(K) \rightarrow C(K)$ be any positive linear isometry with $T|_{A(K)} = \Phi$. Since $1 \in A(K)$ and $\Phi(1) = 1$, we have $T1 = 1$. For each $s \in K$, the functional

$$f \mapsto Tf(s)$$

is a positive norm-one functional on $C(K)$. By the Riesz-Markov representation theorem [2, Theorem 7.17], there is a probability measure μ_s on K such that

$$Tf(s) = \int_K f d\mu_s \quad (f \in C(K)).$$

For every $a \in A(K)$,

$$\int_K a d\mu_s = Ta(s) = \Phi(a)(s) = a(h(s)).$$

Thus μ_s represents the point $h(s)$ on affine functions.

Fix $s \in K$, and choose a peak function $p \in C(K)$ for the point $h(s)$, with $0 \leq p \leq 1$, $p(h(s)) = 1$, and $p < 1$ off $\{h(s)\}$. Since T is an isometry and $\|p\| = 1$, the positive continuous function Tp has norm 1. Compactness gives a point $r \in K$ with

$$1 = Tp(r) = \int_K p d\mu_r.$$

Because $0 \leq p \leq 1$, and $p = 1$ only at $h(s)$, this forces $\mu_r = \delta_{h(s)}$.

On the other hand, μ_r represents $h(r)$ on affine functions, so for every $a \in A(K)$,

$$a(h(r)) = \int_K a d\mu_r = a(h(s)).$$

Continuous affine functions separate points of K , hence $h(r) = h(s)$. Since h is injective, $r = s$. Therefore $\mu_s = \delta_{h(s)}$.

As $s \in K$ was arbitrary, for every $f \in C(K)$ and $s \in K$,

$$Tf(s) = \int_K f d\mu_s = f(h(s)).$$

Thus $T = C_h$, proving uniqueness. $\square$

Corollary 1. *If K is compact metrizable and convex, then for every homeomorphism $h : K \rightarrow K$, the embedding $a \mapsto a \circ h$ from $A(K)$ into $C(K)$ has the unique positive linear isometric extension $f \mapsto f \circ h$. In particular, for $h = \text{id}_K$, the canonical inclusion $A(K) \subseteq C(K)$ has the identity as its unique positive linear isometric extension under this literal $C(K) \rightarrow C(K)$ reading.*

VERIFICATION AND NOVELTY NOTES

The proof is purely analytic; no finite search or numerical verification is used. The only external theorem invoked is the standard Riesz-Markov representation theorem for positive functionals on $C(K)$.

Before promotion, the run's cheap indexes were searched for arXiv:2504.03386, the paper title, and core phrases around order-preserving Hahn-Banach and isometry extensions; no existing packet was found. A bounded web search for exact phrases from the source problem and for Rao's operator-version wording found no direct later answer. Novelty confidence is therefore moderate for the scoped partial theorem, with the main human-review risk being interpretation of the source problem rather than the proof itself.

HUMAN-REVIEW RECOMMENDATION

Review as a scoped partial answer. The proof should be checked for the measure-collapse step and for the stated point-peak hypothesis. The result should not be treated as a full answer to the source problem unless a reviewer accepts the restricted composition-embedding setting as the intended class.

REFERENCES

- [1] Tanmoy Paul and T. S. S. R. K. Rao, *Order-preserving unique Hahn-Banach extensions*, arXiv:2504.03386, 2025. <https://arxiv.org/abs/2504.03386>
- [2] Walter Rudin, *Real and Complex Analysis*, 3rd edition, McGraw-Hill, 1987.